# JEAN COTE AGCM, AB, Canada

WMO: **715410**

Lat: **55.9128N** Lon: **117.1197W** Elev: **638** StdP: **93.89** Time Zone: **-7.00 (NAM)** Period: **08-19** WBAN: **99999**

## Annual Heating, Humidification, and Ventilation Design Conditions

| | Coldest Month | Heating DB | | Humidification DP/MCDB and HR | | | | | | Coldest Month WS/MCDB | | | | MCWS/PCWD to 99.6% DB | | WSF | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 99.6% | | | 99% | | | 0.4% | | 1% | | | | | |
| | | 99.6% | 99% | DP | HR | MCDB | DP | HR | MCDB | WS | MCDB | WS | MCDB | MCWS | PCWD | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (1) | **12** | **-30.3** | **-27.7** | **-33.9** | **0.2** | **-30.3** | **-31.1** | **0.2** | **-27.6** | **11.4** | **1.2** | **10.1** | **-0.7** | **2.7** | **210** | **0.677** | (1) |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| | Hottest Month | Hottest Month DB Range | Cooling DB/MCWB | | | | | | Evaporation WB/MCDB | | | | | | MCWS/PCWD to 0.4% DB | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0.4% | | 1% | | 2% | | 0.4% | | 1% | | 2% | | | | |
| | | | DB | MCWB | DB | MCWB | DB | MCWB | WB | MCDB | WB | MCDB | WB | MCDB | MCWS | PCWD | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (2) | **7** | **11.8** | **27.7** | **16.3** | **25.7** | **15.6** | **24.1** | **15.1** | **18.2** | **24.5** | **17.2** | **23.6** | **16.2** | **22.1** | **3.5** | **250** | (2) |

| | Dehumidification DP/MCDB and HR | | | | | | | | | Enthalpy/MCDB | | | | | | Extreme Max WB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0.4% | | | 1% | | | 2% | | | 0.4% | | 1% | | 2% | | | |
| | DP | HR | MCDB | DP | HR | MCDB | DP | HR | MCDB | Enth | MCDB | Enth | MCDB | Enth | MCDB | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (3) | **15.8** | **12.1** | **20.9** | **14.7** | **11.3** | **19.8** | **13.7** | **10.6** | **18.5** | **54.0** | **24.2** | **50.8** | **23.8** | **47.8** | **22.1** | **21.7** | (3) |

## Extreme Annual Design Conditions

| | Extreme Annual WS | | | | Extreme Annual Temperature | | | | n-Year Return Period Values of Extreme Temperature | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Mean | | Standard Deviation | | n=5 years | | n=10 years | | n=20 years | | n=50 years | | |
| | 1% | 2.5% | 5% | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| | (a) | (b) | (c) | | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (4) | **9.5** | **8.4** | **7.5** | DB | **-34.2** | **30.1** | **2.5** | **1.5** | **-36.0** | **31.2** | **-37.4** | **32.1** | **-38.8** | **33.0** | **-40.6** | **34.1** | (4) |
| (5) | | | | WB | **-34.3** | **19.7** | **2.4** | **1.6** | **-36.1** | **20.9** | **-37.5** | **21.8** | **-38.9** | **22.7** | **-40.6** | **23.9** | (5) |

## Monthly Climatic Design Conditions

| | | | | Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (6) | **Temperatures, Degree-Days and Degree-Hours** | DBAvg | | **2.5** | **-10.9** | **-11.2** | **-5.4** | **2.9** | **10.7** | **14.5** | **16.4** | **15.5** | **10.6** | **3.5** | **-5.9** | **-11.9** | (6) |
| (7) | | DBStd | | **12.13** | **8.60** | **8.25** | **7.78** | **5.10** | **4.56** | **3.39** | **2.83** | **3.48** | **4.53** | **4.87** | **6.51** | **7.54** | (7) |
| (8) | | HDD10.0 | | **3401** | **649** | **592** | **478** | **218** | **47** | **5** | **0** | **3** | **46** | **209** | **476** | **678** | (8) |
| (9) | | HDD18.3 | | **5821** | **907** | **826** | **736** | **462** | **237** | **122** | **74** | **100** | **233** | **461** | **726** | **937** | (9) |
| (10) | | CDD10.0 | | **655** | **0** | **0** | **0** | **6** | **69** | **139** | **197** | **173** | **64** | **7** | **0** | **0** | (10) |
| (11) | | CDD18.3 | | **34** | **0** | **0** | **0** | **0** | **2** | **6** | **13** | **12** | **1** | **0** | **0** | **0** | (11) |
| (12) | | CDH23.3 | | **531** | **0** | **0** | **0** | **4** | **48** | **96** | **178** | **168** | **37** | **0** | **0** | **0** | (12) |
| (13) | | CDH26.7 | | **94** | **0** | **0** | **0** | **1** | **9** | **14** | **29** | **38** | **4** | **0** | **0** | **0** | (13) |
| (14) | **Wind** | WSAvg | | **3.6** | **4.0** | **3.7** | **3.6** | **3.9** | **4.0** | **3.5** | **3.0** | **3.1** | **3.4** | **3.8** | **3.7** | **3.7** | (14) |
| (15) | **Precipitation** | PrecAvg | | **458** | **30** | **21** | **19** | **18** | **43** | **71** | **69** | **58** | **38** | **26** | **30** | **27** | (15) |
| (16) | | PrecMax | | **664** | **84** | **42** | **40** | **36** | **108** | **141** | **104** | **120** | **117** | **56** | **64** | **61** | (16) |
| (17) | | PrecMin | | **274** | **9** | **7** | **3** | **2** | **13** | **31** | **15** | **20** | **2** | **9** | **9** | **5** | (17) |
| (18) | | PrecStd | | **95** | **21** | **9** | **10** | **10** | **27** | **31** | **30** | **28** | **26** | **12** | **18** | **14** | (18) |
| (19) | **Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures** | 0.4% | DB | | **5.7** | **5.8** | **10.1** | **22.2** | **28.0** | **28.7** | **29.6** | **30.4** | **26.9** | **20.9** | **10.9** | **4.5** | (19) |
| (20) | | | MCWB | | **2.7** | **2.5** | **5.4** | **10.6** | **13.2** | **16.2** | **18.5** | **17.8** | **15.9** | **11.9** | **6.7** | **1.6** | (20) |
| (21) | | 2% | DB | | **4.5** | **3.7** | **7.8** | **16.9** | **24.2** | **25.8** | **27.1** | **27.4** | **24.0** | **15.8** | **6.6** | **2.5** | (21) |
| (22) | | | MCWB | | **2.0** | **1.4** | **3.5** | **7.5** | **11.5** | **14.7** | **16.6** | **16.9** | **14.3** | **9.3** | **3.7** | **0.4** | (22) |
| (23) | | 5% | DB | | **3.1** | **2.0** | **5.7** | **13.4** | **21.8** | **24.0** | **25.3** | **25.0** | **21.3** | **12.8** | **4.4** | **0.9** | (23) |
| (24) | | | MCWB | | **1.2** | **0.1** | **2.1** | **6.1** | **10.6** | **14.2** | **16.4** | **15.9** | **13.4** | **7.2** | **2.1** | **-1.1** | (24) |
| (25) | | 10% | DB | | **1.3** | **0.0** | **4.0** | **10.7** | **19.8** | **22.1** | **23.6** | **23.0** | **18.4** | **10.6** | **2.3** | **-1.2** | (25) |
| (26) | | | MCWB | | **-0.4** | **-1.5** | **1.0** | **4.6** | **9.7** | **13.3** | **15.7** | **15.4** | **11.7** | **6.1** | **0.6** | **-2.7** | (26) |
| (27) | **Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures** | 0.4% | WB | | **3.0** | **2.8** | **5.7** | **11.0** | **14.2** | **17.9** | **20.0** | **19.2** | **16.8** | **12.6** | **6.7** | **1.9** | (27) |
| (28) | | | MCDB | | **5.2** | **5.5** | **9.8** | **21.9** | **24.3** | **25.7** | **26.8** | **26.6** | **25.4** | **19.0** | **10.3** | **4.0** | (28) |
| (29) | | 2% | WB | | **2.2** | **1.6** | **3.8** | **8.3** | **12.7** | **16.4** | **18.6** | **18.0** | **15.1** | **9.8** | **4.1** | **0.5** | (29) |
| (30) | | | MCDB | | **4.2** | **3.5** | **7.4** | **16.2** | **21.1** | **23.2** | **24.4** | **24.8** | **22.0** | **15.2** | **6.4** | **2.4** | (30) |
| (31) | | 5% | WB | | **1.3** | **0.2** | **2.3** | **6.3** | **11.5** | **15.2** | **17.5** | **17.1** | **13.7** | **7.9** | **2.2** | **-0.9** | (31) |
| (32) | | | MCDB | | **3.1** | **2.0** | **5.3** | **12.4** | **20.0** | **21.6** | **23.2** | **23.6** | **20.0** | **11.7** | **3.9** | **0.9** | (32) |
| (33) | | 10% | WB | | **-0.4** | **-1.6** | **1.2** | **4.9** | **10.5** | **14.0** | **16.5** | **16.1** | **12.4** | **6.5** | **0.7** | **-2.7** | (33) |
| (34) | | | MCDB | | **1.2** | **0.0** | **3.8** | **10.2** | **18.2** | **20.5** | **22.1** | **21.7** | **17.8** | **10.1** | **2.1** | **-1.3** | (34) |
| (35) | **Mean Daily Temperature Range** | | MDBR | | **8.0** | **8.3** | **9.3** | **10.3** | **12.8** | **11.3** | **11.8** | **12.2** | **11.5** | **8.9** | **7.3** | **7.3** | (35) |
| (36) | | 5% DB | MCDBR | | **8.0** | **8.4** | **10.0** | **14.6** | **16.0** | **14.5** | **14.7** | **14.9** | **15.6** | **12.4** | **9.0** | **8.9** | (36) |
| (37) | | | MCWBR | | **6.5** | **6.7** | **6.6** | **7.8** | **6.8** | **6.1** | **6.8** | **6.4** | **7.2** | **7.1** | **6.5** | **7.7** | (37) |
| (38) | | 5% WB | MCDBR | | **7.7** | **8.0** | **9.6** | **13.7** | **14.6** | **12.6** | **13.1** | **13.7** | **14.6** | **11.8** | **8.5** | **9.1** | (38) |
| (39) | | | MCWBR | | **6.3** | **6.5** | **6.5** | **7.8** | **6.6** | **5.9** | **6.9** | **6.6** | **7.2** | **7.1** | **6.4** | **7.8** | (39) |
| (40) | **Clear-Sky Solar Irradiance** | taub | | | **0.235** | **0.245** | **0.284** | **0.326** | **0.339** | **0.347** | **0.349** | **0.346** | **0.307** | **0.281** | **0.241** | **0.227** | (40) |
| (41) | | taud | | | **2.301** | **2.353** | **2.346** | **2.377** | **2.407** | **2.407** | **2.427** | **2.414** | **2.532** | **2.563** | **2.431** | **2.223** | (41) |
| (42) | | Ebn at Noon | | | **715** | **857** | **893** | **897** | **902** | **896** | **887** | **866** | **863** | **809** | **715** | **618** | (42) |
| (43) | | Edh at Noon | | | **49** | **69** | **91** | **103** | **107** | **109** | **104** | **99** | **76** | **58** | **43** | **41** | (43) |
| (44) | **All-Sky Solar Radiation** | RadAvg | | | **0.73** | **1.72** | **3.28** | **4.70** | **5.61** | **6.01** | **5.87** | **4.77** | **3.25** | **1.74** | **0.78** | **0.50** | (44) |
| (45) | | RadStd | | | **0.07** | **0.15** | **0.27** | **0.30** | **0.42** | **0.47** | **0.38** | **0.39** | **0.26** | **0.21** | **0.06** | **0.04** | (45) |

## Historical Trends

| | | DBAvg | Heating | | Cooling | | | Degree-Days | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 99% DB | 99% DP | 1% DB | 1% WB | 1% DP | HDD10.0 | HDD18.3 | CDD10.0 | CDD18.3 | |
| | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | |
| (46) | Station Only | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (46) |
| (47) | Regional (20 neighbors) | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **+57** | **N/A** | (47) |

Nomenclature: See separate page